

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 12)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

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0 1 . 2

Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

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0 1 . 3

Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]

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02 . 1

Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

02 . 2

Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

02 . 3

Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

0 2 . 4

Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

10

0 3 . 1

Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

0 3 . 2

Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

03 . 3

Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

10

04

Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

04 . 1

State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

04 . 2

Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

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0 4 . 3 Describe **two** effects of acid rain on non-living objects. **[2 marks]**

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0 4 . 4 Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere. **[2 marks]**

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0 5 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box. **[1 mark]**

- | | |
|------------------------------|--------------------------|
| A Its total annual discharge | ☐ |
| B The albedo of its surface | ☐ |
| C Its salinity | ☐ |
| D The age of its bedrock | ☐ |

05 . 2

Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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05 . 3

Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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06 . 1

Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

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06 . 2

Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

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0 6 . 3

Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

10

0 7

Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 7 . 1

Define the term **environmental half-life**.

[1 mark]

0 7 . 2

Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

07 . 3

Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

07 . 4

Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

07 . 5

Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

1

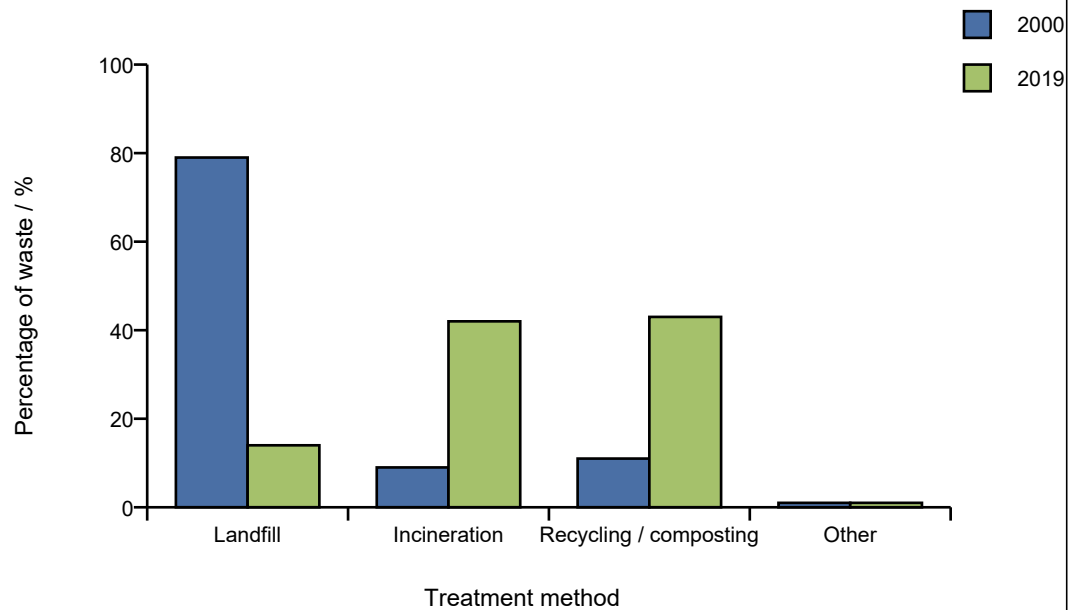
2

10

08

Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



08

1

Describe the changes shown in Figure 1.

[2 marks]

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08

2

Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

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08

3

Describe **four** features of a well-managed landfill site.

[4 marks]

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0 8 . **4** Suggest why incineration has increased more than recycling for some waste types. **[1 mark]**

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10

0 9 . **1** Complete Table 1 to show one method of controlling each source of railway noise. **[4 marks]**

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

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09 . 2

Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

09 . 3

Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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09 . 4

Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

Extra space

10 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

10

Write an essay on one of the following topics.

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Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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END OF QUESTIONS